

Progressive gate-unfreezing of the QFT operator

One gate thawed per stage, trained to a plateau — three unfreeze orderings $\times$ two initialisations on DIV2K-8q

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Question

The **QFTBasis(8, 8)** (72 $U(2)$ gates) starts either at the QFT-family **identity** ($T_{t=0}(x) = x$) or **Haar-random**. We **unfreeze one gate per stage**, train it to a plateau, then thaw the next. Does the **order** — block-growth (bg), left $\rightarrow$ right (lr), right $\rightarrow$ left (rl) — or the **initialisation** change the trajectory or the endpoint?

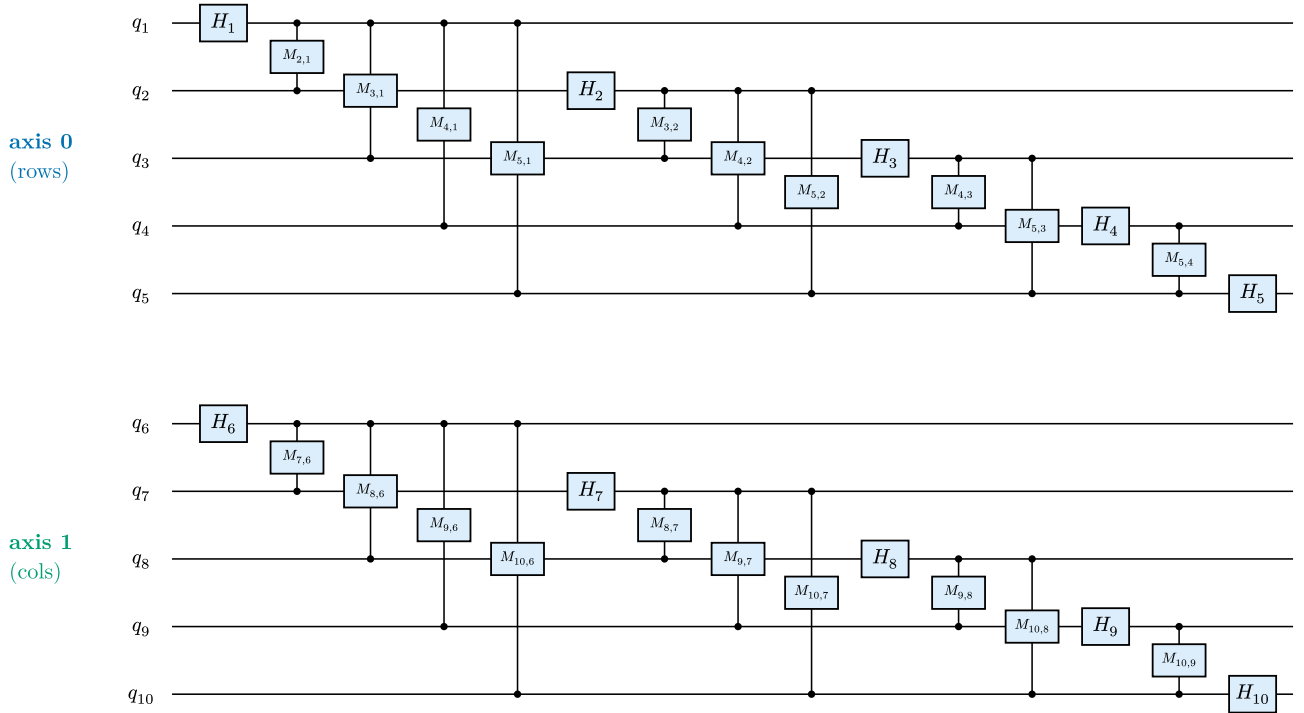


Figure 1: The **2-D QFT(5,5)** — two independent QFT(5) blocks, one per image axis (**axis 0** = $q_1..q_5$, rows; **axis 1** = $q_6..q_{10}$, columns). H_i = Hadamard on q_i ; $M_{j,i}$ = controlled-phase between q_j and q_i ($j > i$). The DIV2K runs use the analogous QFT(8,8) (axis 1 = $q_9..q_{16}$, 72 gates).

Every ordering thaws the gates of Figure 1 one at a time, on **both** axes:

- **block-growth (bg)**: grow a **square $2^k \times 2^k$ block** — block-stage k thaws qubit k on both axes. H_1, H_6 ; then $H_2, M_{2,1}, H_7, M_{7,6}$; then $H_3, M_{3,1}, M_{3,2}, H_8, M_{8,6}, M_{8,7}$; ... up to $k\{=\}5$ ($H_5, ..., M_{5,4}, H_{10}, ..., M_{10,9}$). Long-range couplings last. (At $m = 8$ the axis-1 partner of H_1 is H_9 — hence the early H_9 in the dynamics figure.)
- **left $\rightarrow$ right (lr)**: QFT construction order, axis 0 then axis 1 — $H_1, M_{2,1}, ..., H_5$, then $H_6, M_{7,6}, ..., H_{10}$.
- **right $\rightarrow$ left (rl)**: the reverse — H_{10} first, H_1 last.

Setup

Fixed batch of 50 training images; **MSELoss** on the top-10% coefficients. pdft's JIT Adam step runs with a per-stage frozen set (moments reset per stage); a Riemannian grad-norm probe gives the plateau signal. A stage ends at $\|g\| < 10^{-5}$ or $|\Delta L| < 10^{-5}$ (min 5 steps), else an 800-step cap. Random init: Haar $U(2)$ per Hadamard, uniform phase per controlled-phase, seed shared across orderings.

Training dynamics

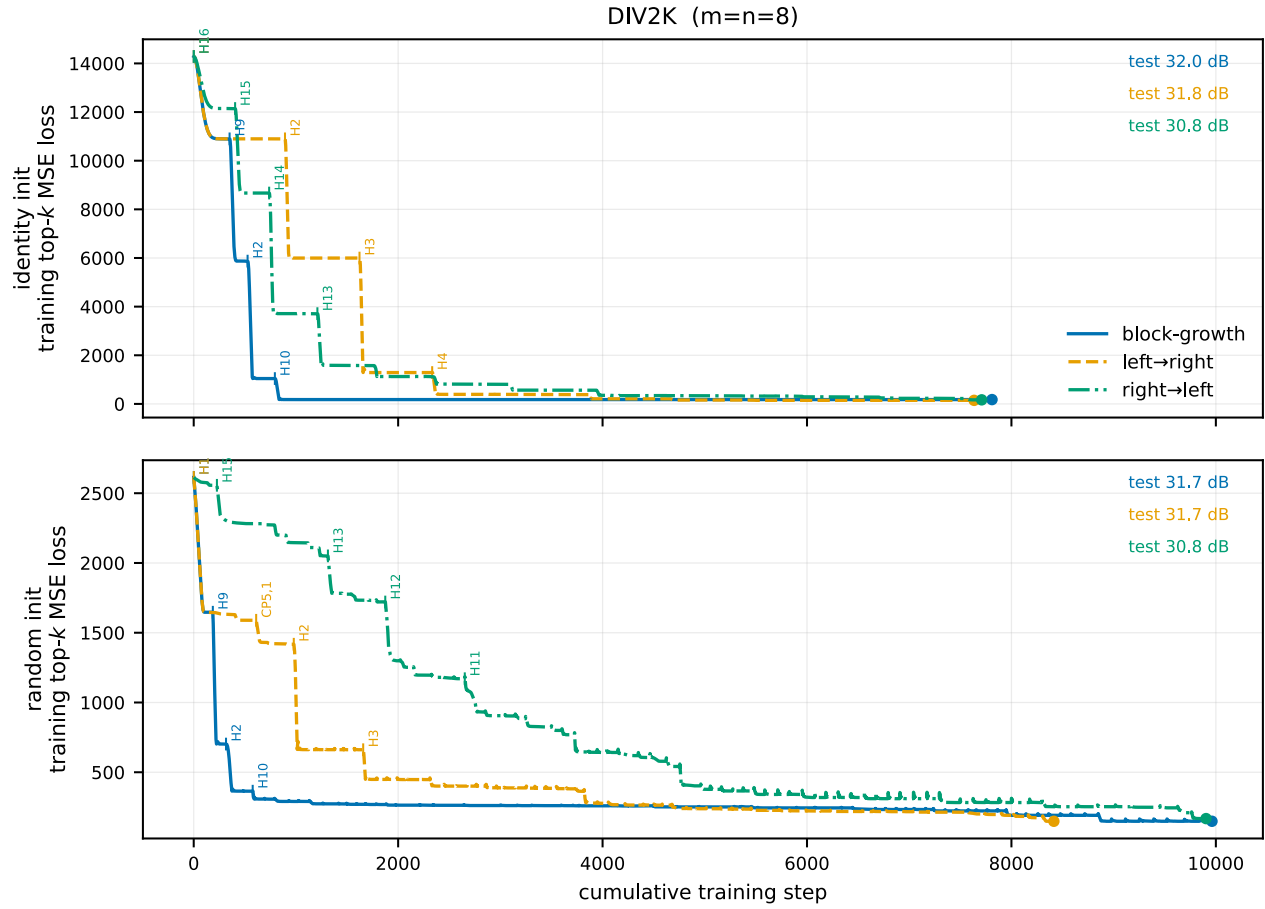


Figure 2: Absolute **training** top- k MSE loss vs cumulative step, one curve per ordering (rows: identity / random init). Tick labels mark **which gate** (e.g. H7, CP3,1) was thawed at each of the largest drops. The y-axis is the **training** top-10%-coefficient MSE loss; endpoints are annotated with the separate **test PSNR@ $\rho=0.20$** (image reconstruction on held-out data) — the two are different metrics, so the curve’s height does not determine the PSNR. Per-cell loss+grad-norm staircases: [div2k_8q/figures/](#).

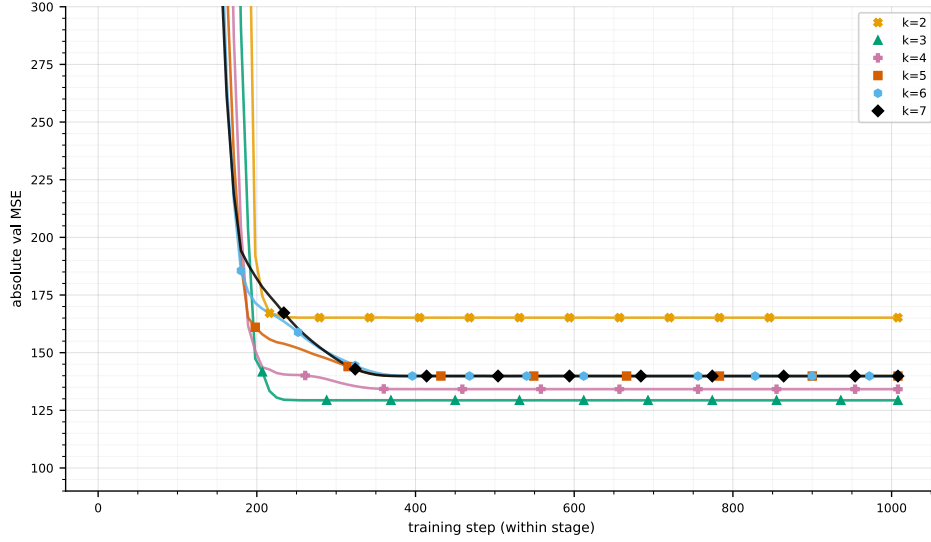


Figure 3: **Block-size** curriculum (qft_progressive, DIV2K) for comparison: per-stage validation MSE for block sizes $k = 2..7$, each one whole $2^k \times 2^k$ QFT trained 1008 steps. Eight whole-block stages here vs the 72 one-gate-at-a-time stages above — same ≈ 31.7 dB endpoint (table below).

Reconstruction PSNR (DIV2K)

Gate-unfreeze endpoints **together with** the qft_progressive block-size baseline (train all $k(k+1)$ gates of a $2^k \times 2^k$ QFT at once, per stage k). DIV2K test PSNR (dB) at four keep ratios ρ :

configuration	$\rho\{=\}.05$	$\rho\{=\}.10$	$\rho\{=\}.15$	$\rho\{=\}.20$	train MSE
unfreeze · identity · bg	19.4	26.9	29.7	32	178.2
unfreeze · identity · lr	25.2	27.9	30	31.8	146.6
unfreeze · identity · rl	24.8	27.2	29.1	30.8	167.8
unfreeze · random · bg	25.2	27.8	29.8	31.7	148.6
unfreeze · random · lr	25.2	27.8	29.8	31.7	148.6
unfreeze · random · rl	24.8	27.2	29.1	30.8	167.8
block-size · k=1 (2×2)	9.4	12.5	16.2	21.1	—
block-size · k=2 (4×4)	19.1	26.9	29.8	32	—
block-size · k=3 (8×8)	25.2	28.1	30.3	32.3	—
block-size · k=4 (16×16)	25.3	28	30.1	32	—
block-size · k=5 (32×32)	25.2	27.8	29.8	31.7	—
block-size · k=6 (64×64)	25.2	27.8	29.8	31.7	—
block-size · k=7 (128×128)	25.2	27.8	29.8	31.7	—
block-size · k=8 (256×256)	25.2	27.8	29.8	31.7	—
classical · block-DCT 8×8	26.1	29.4	31.9	34	—
classical · block-FFT 8×8	24.5	27.1	29.1	30.8	—

train MSE = final top-10%-coefficient training loss (absolute, train batch) — a different metric from the PSNR columns, shown per gate-unfreeze run; not defined for the block-size / classical references.

All learned QFT schemes reach the **same** QFT(8,8) endpoint, ≈ 31.7 dB @ $\rho\{=\}.20$: the block-size curriculum saturates by $k = 2$ and the gate-unfreeze sweep lands there from either init and any order — **schedule-independent** (rl trails ≈ 1 dB; every gate-unfreeze stage ended on the loss- Δ plateau, none at the step cap). For reference, the best **any** prior trained basis reaches is 33.7 dB (rich/identity). The classical **block-DCT 8×8** is in fact the strongest here (34 dB @ $\rho\{=\}.20$), ahead of every learned QFT basis; **block-FFT 8×8** is weaker (30.8 dB) — the QFT-family operators sit between the two classical block transforms.

Is the fixed top-10% training objective a confound? The classical transforms are inherently rate-matched (keep top- ρ of a **fixed** operator at every ρ), while the learned QFT trains once at top-10% and is scored at 5–20%. This does not explain its deficit: (i) at the **matched** rate $\rho\{=\}.10$ the classical block-DCT (29.4 dB) still beats the best gate-unfreeze (27.9 dB); and (ii) retraining the QFT with top- k matched to each ρ raises the matched-rate PSNR by ≤ 0.07 dB (controlled all-at-once probe, `reference/rate_matched_div2k.json`) — the operator’s energy compaction is essentially rate-agnostic, so the top- k choice is not the issue.

Seed robustness (random init). Across **17** random-init seeds (block-growth), **16 land at 31.66 dB** @ $\rho\{=\}.20$ (mean 31.64, $\sigma = 0.09$, max 31.66); a single seed dipped to 31.29. So the random endpoint is seed-invariant and **never exceeds** the 31.66 attractor — the optimum is a stable basin, not a lucky draw, and reseeding cannot reach the identity result (31.98) or the 8×8 -block 32.26 (`reference/random_seed_sweep_div2k.json`, $n = 17$).

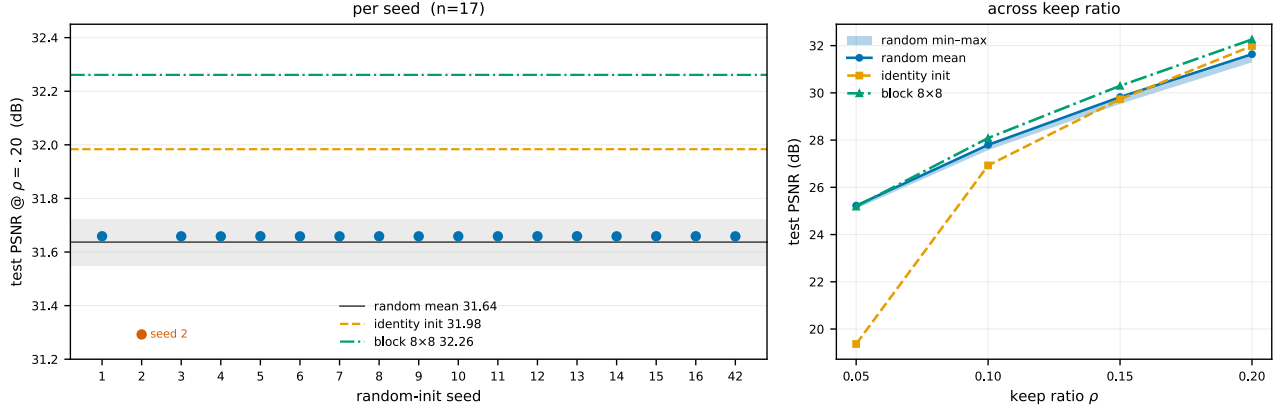


Figure 4: **Random-init seed sweep** (block-growth, DIV2K, $n = 17$). **Left:** every seed’s test PSNR@ $\rho\{=\}.20$: 16 sit on the 31.66-dB attractor (grey band = mean $\pm \sigma$) and the lone outlier (seed 2) at 31.29 — none reach the identity-init (31.98) or 8×8 -block (32.26) reference. **Right:** the random min-max band is negligible at every ρ — yet identity init is **worse** at low ρ (19.4 dB @ $\rho\{=\}.05$ vs 25.2), crossing above only near $\rho\{=\}.20$.

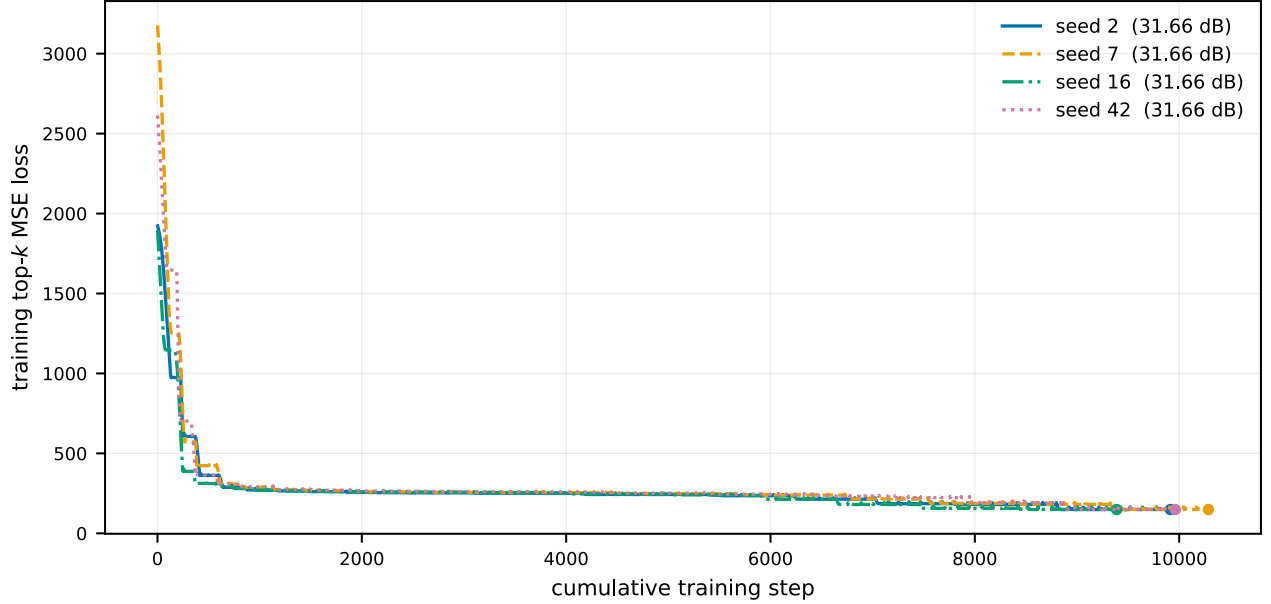


Figure 5: **Per-seed training dynamics** (random init, block-growth, DIV2K): one absolute training top- k MSE curve per seed (2, 7, 16, 42). The seeds start from different Haar inits (hence different L_0) and descend along visibly different early paths, yet all converge to the same ≈ 31.66 dB attractor by step ≈ 1000 . **Caveat:** on this rerun (grad-probe every 5 steps) seed 2 **itself** reached the attractor (31.66 dB), not the 31.29-dB outlier recorded in the $n = 17$ sweep above — the lone-outlier result is **not robust** to the grad-probe cadence / device floating-point, so the seed 2 dip should be read with caution. Data: `reference/random_seed_dynamics_div2k.json`.

Reading

The staircase shows each gate’s marginal value: the big drops are the **early** gates (the marked Hadamards, e.g. H9, H2 for block-growth); later gates only fine-tune. One caveat on the endpoint labels — the train-loss order need not match the test-PSNR order (**bg** has the **highest** train MSE yet the **best** test PSNR): the loss is a top-10%-coefficient proxy on the train batch, PSNR a 20%-keep reconstruction on the test set.